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1 考点 1: 回归对冲和主成分分析

1. Assume that a trader is making a relative value trade, selling a U.S. Treasury bond and correspondingly purchasing a U.S. Treasury TIPS. Based on the current spread between the two securities, the trader shorts \$120 million of the nominal bond and purchases \$105.6 million of TIPS. The trader then starts to question the amount of the hedge due to changes in yields on TIPS in relation to nominal bonds. He runs a regression and determines from the output that the nominal yield changes by 1.0352 basis points per basis point change in the real yield. Would the trader adjust the hedge, and if so, by how much?

- A. No.
- B. Yes, by \$3.12 million (purchase additional TIPS)
- C. Yes, by \$3.15 million (sell a portion of the TIPS)
- D. Yes, by \$2.88 million (Purchase additional TIPS)

答案:B

解析:

步骤 1: 理解对冲条件

- 完全对冲需要: 名义债券的美元久期变化 = TIPS 的美元久期变化
- 即: $MD_n \times P_n \times \Delta y_n = MD_r \times P_r \times \Delta y_r$

步骤 2: 原始对冲假设

- 原始头寸: 卖空 \$120M 名义债券, 买入 \$105.6M TIPS
- 原始假设: $\Delta y_n = \Delta y_r$ (名义收益率变化等于实际收益率变化)
- 因此: $MD_n \times 120 = MD_r \times 105.6$

步骤 3: 根据回归结果调整

- 回归显示: $\Delta y_n = 1.0352 \times \Delta y_r$
- 新的对冲条件:

$$MD_n \times 120 \times \Delta y_n = MD_r \times P_r \times \Delta y_r$$

- 代入 $\Delta y_n = 1.0352 \times \Delta y_r$:

$$MD_n \times 120 \times 1.0352 \times \Delta y_r = MD_r \times P_r \times \Delta y_r$$

- 化简: $MD_n \times 120 \times 1.0352 = MD_r \times P_r$

步骤 4: 计算新的 TIPS 头寸

- 从步骤 2: $MD_n = \frac{105.6}{120} \times MD_r$
- 代入步骤 3 的结果:

$$\frac{105.6}{120} \times MD_r \times 120 \times 1.0352 = MD_r \times P_r$$

- 化简: $105.6 \times 1.0352 = P_r$
- 因此: $P_r = 108.72$ million

步骤 5: 计算调整金额

- 需要增加的 TIPS 头寸: $108.72 - 105.6 = 3.12$ million
- 因此需要额外购买 \$3.12M 的 TIPS。

考察: 掌握回归对冲 (根据收益率变化关系调整对冲比例)

2. If a trader is creating a fixed income hedge, which hedging methodology would be least effective if the trader is concerned about the dispersion of the change in the corporate bond yield for a particular change in the government bond yield?

- A. One-variable regression hedge.
- B. DV01 hedge
- C. Two-variable regression hedge.
- D. Principal components hedge.

答案:B

解析:

A 选项错误。

- 单变量回归对冲可以捕捉公司债和政府债收益率变化之间的关系。

B 选项正确。

- DV01 对冲假设公司债和政府债的收益率变化幅度相同，无法处理两者之间的分散性。具体来说：
 - DV01 对冲基于收益率平行移动的假设
 - 无法反映信用利差的变化
 - 忽略了收益率曲线的非平行移动

C 选项错误。

- 双变量回归对冲可以更好地捕捉收益率变化的分散性。

D 选项错误。

- 主成分对冲可以处理收益率曲线的各种变化形态。

考察： 理解对冲方法（DV01 对冲无法处理收益率分散性）